

GEN-3 SAFE HELICOPTER

COMPREHENSIVE ENGINEERING CONCEPT DESIGN DESCRIPTION

Revision 02 – Full Expanded Edition • 2026

Founder

Amirbahador Atai

Document Status

Expanded Concept / Pre-Feasibility / Systems Engineering Baseline

Important Status Statement. *This document is a concept engineering baseline. It records the agreed architecture, preliminary sizing, safety philosophy, assumptions, interfaces, failure cases, verification strategy and open engineering items. It is **not** a type certificate, airworthiness approval, production drawing package, flight manual, or proof that the aircraft can presently meet every target. All certification-critical claims require formal analysis, testing and authority acceptance.*

Prepared as a consolidated master document so that future revisions can replace assumptions and TBD values with traceable calculations, supplier data, test results and certification evidence. This Rev 02 Full Expanded Edition significantly increases content depth, adds detailed discussion of each major subsystem, expands the safety and certification framework, and provides a more complete development and risk picture relative to earlier revisions.

01. Document Control & Configuration

Field	Baseline
Document	GEN-3 SAFE HELICOPTER — Comprehensive Engineering Concept Design Description
Revision	02 – Full Expanded Edition
Year	2026
Founder	Amirbahador Atai
Status	Concept / Pre-feasibility / Systems Engineering Baseline
Certification	Not certified — concept only
Production	Not production released
Numerical status	Targets / preliminary unless explicitly verified
Development assurance	Aligned with SAE ARP4754A principles (FAA AC 20-174)
Language	English
Classification	Confidential – Concept Baseline

This revision expands the original baseline by refining requirements traceability, clarifying preliminary sizing assumptions, expanding the safety analysis framework, improving the emergency scenario matrix, adding clearer verification and certification pathway guidance, and providing a more complete development roadmap and risk register. All numerical values remain conceptual targets pending detailed analysis and test validation. The document is intended for use by the founder, potential technical partners, investors evaluating early-stage architecture, and future certification teams as a starting point for formal requirements capture.

Configuration control: This is the master concept baseline. Any future numerical change, architectural change or requirement change should be recorded with a clear rationale, source of data (analysis, test, supplier) and impact assessment on mass, power, safety and certification path.

02. Executive Definition & Design Intent

GEN-3 is a heavy modular amphibious VTOL aircraft concept centered on an independently survivable Safety Capsule. The architecture prioritizes layered survivability: prevent escalation, isolate the failure, retain control, separate the capsule when required, absorb impact energy if separation fails, stabilize on water, preserve flotation, and provide multiple escape routes. The concept is deliberately different from conventional single-main-rotor or tandem-rotor helicopters. It trades classical simplicity for distributed redundancy, modular isolation and a last-line occupant survival volume that can function after major airframe degradation.

The design is intended for heavy logistics, disaster response, offshore and maritime support, and specialized rescue roles where the combination of high lift, water operation and extreme survivability has operational value. It is not optimized for low-cost light utility flying; it is optimized for high-consequence missions in which the cost of a catastrophic event is very high.

- Eight distributed primary lift fans (~18 m diameter each) and six capsule emergency fans (~10 m diameter each); total conceptual fan count: 14.
- Mission endurance target: at least 3 hours plus reserve.
- Reference MTOW target region: approximately 150–160 t (no hard 100 t cap).
- Reference thrust-to-weight: ~1.5 for design operation; ~1.7 for maximum/emergency sizing.
- Initial fuel planning target: approximately 30 t Jet-A / Jet-A1.
- Primary structural material target: ~70% CFRP by primary structural mass.
- Central detachable, buoyant, crashworthy Safety Capsule with independent power, life support, emergency fans, parachutes and multi-directional egress.
- Individual aircraft modules can be isolated and, where required, released in a controlled jettison sequence.
- Capsule-to-airframe interfaces use multiple springs, dampers and progressive energy absorbers so the capsule remains protected if separation fails.
- Water takeoff, water landing, flotation, capsizing tolerance / self-righting and inverted-state egress are explicit design objectives.

Safety Philosophy: Layered survivability. Critical functions have independent or segregated paths where practical. Fire is contained before becoming a structural or occupant threat. Energy sources are isolated from the occupant survival volume. Emergency functions require deterministic trigger conditions and testable outcomes. The design does not rely on a single “perfect” system; it relies on progressive containment, isolation, separation and absorption so that a single major failure does not automatically cascade into total loss of the occupant volume.

03. Mission, Operating Concept & Design Philosophy

Primary Mission Roles

Heavy logistics and cargo support; disaster response (earthquake, flood, fire, industrial accident); offshore / maritime logistics and rescue; remote-area transport where runways or prepared helipads are unavailable; specialized rescue and recovery operations including inverted or water-immersed recovery scenarios. Secondary roles may include support to large construction projects, heavy external-load work in limited-access areas, and government / civil-protection rapid response.

Mission Sequence

Pre-flight health check (including module isolation readiness, capsule systems, flotation and egress) → ground or water takeoff → climb / transition → cruise / mission phase → reserve → descent → ground or water landing → stabilization / evacuation if required. The sequence is deliberately written so that water operation is a first-class capability, not an afterthought.

Core Design Principles

- Critical functions should have independent or segregated paths where practical.
- Fire should be contained before becoming a structural or occupant threat.
- Energy sources should be isolated from the occupant survival volume.
- Emergency functions require deterministic trigger conditions and testable outcomes.
- Modular architecture enables isolation and controlled release of failed modules.
- The Safety Capsule remains the last line of protection for occupants under all credible failure modes.
- Water capability and inverted-state survival are treated as design requirements, not optional features.
- Every major safety claim must eventually map to a verification method (analysis, inspection, test or demonstration).

The operating concept assumes that the aircraft will sometimes operate in environments where external rescue is delayed or impossible. Therefore independent life support, flotation and multi-orientation egress are not “nice to have”; they are central to the value proposition of the Safety Capsule.

04. System Requirements (Selected)

The following table is a selected extract of the requirements baseline. Each requirement carries an identifier, a concise statement of the target, and a high-level verification approach. Full requirements traceability (requirement → architecture element → analysis / test → evidence) will be developed in later systems-engineering phases under ARP4754A-aligned processes.

ID	Requirement / Target	Status / Verification
SYS-001	8 primary lift fans, ~18 m each	Architecture target / Inspection + analysis
SYS-002	6 capsule fans, ~10 m each	Architecture target / Inspection + analysis
SYS-003	≥ 3 h mission endurance + reserve	Mission target / Simulation + flight test
SYS-004	T/W ~1.5 design; ~1.7 max/emergency	Sizing target / Analysis + test
SYS-005	~150–160 t MTOW region	Mass target / Mass statement closure
SYS-006	~30 t initial fuel (Jet-A/Jet-A1)	Fuel target / Mission simulation
SYS-007	~70% CFRP primary structural mass	Materials target / Mass fraction report
CAP-001	Detachable central Safety Capsule	Architecture / Design review + separation test
CAP-002	Independent capsule power & life support	Safety / Power-loss test
CAP-003	Multi-point spring/damper crash isolation	Safety / FEA + dynamic test
CAP-004	Multi-orientation emergency egress	Safety / Evacuation test
CAP-005	Water landing and flotation	Safety / Hydrodynamic + flight test
CAP-006	Emergency fans + parachute recovery	Safety / Drop / flight test
MOD-001	Individual module isolation & controlled release	Architecture / Ground / integrated test
MOD-002	Fire-driven jettison after unsuccessful containment	Safety concept / Fire + jettison test
CTRL-001	Redundant flight control (triple conceptual)	Safety target / System verification
FUEL-001	Crashworthy isolated fuel system	Safety / Impact + fire test
WAT-001	Water-impact survival	Safety / Impact test
WAT-002	Capsize-tolerant / self-righting capsule target	Safety / Pool / wave test
PWR-001	Segmented HV distribution with fault isolation	Architecture / Fault injection test
THR-001	Thermal management for continuous high-power hover	Analysis + thermal test

Additional requirements covering noise, vibration, human factors, maintainability, dispatch reliability, ground handling and production quality will be captured in later revisions once the primary architecture is frozen and the certification basis is agreed with the authority.

05. Overall Aircraft Architecture

The aircraft is organized around a central occupant Safety Capsule. Eight distributed primary lift / power modules provide main lift and control. Six emergency capsule fans are arranged around / below the capsule for post-separation recovery. Protected central fuel architecture, segmented high-voltage electrical distribution, modular primary composite structure with metallic / hybrid hard points, independent capsule systems, dedicated water-contact / flotation / recovery architecture, and dedicated emergency separation and jettison interfaces complete the baseline architecture.

The physical arrangement is intended to keep the highest-energy sources (fuel, high-power generation, primary lift fans) outside the immediate survival volume of the capsule while still allowing structural load paths that can carry normal flight, asymmetric and crash loads. Module interfaces are designed so that a failed module can be electrically, thermally and mechanically isolated without compromising the remaining aircraft or the capsule.

05.1 Fan Disk Areas (Geometric Reference)

For a circular fan, $A = \pi D^2/4$. An 18 m fan has approximately 254.47 m² disk area; eight provide approximately 2,035.75 m². A 10 m fan has approximately 78.54 m²; six provide approximately 471.24 m². Combined geometric area is approximately 2,506.99 m². Geometric disk area is not useful thrust by itself. Actual lift depends on disk loading, induced velocity, efficiency, blade geometry, air density, tip speed, power and control margin. Capsule fans are not automatically credited as normal primary lift; they are sized and credited for the separated-capsule emergency case.

Group	Qty	Diameter	Area / fan	Combined
Main	8	18 m	~254.47 m ²	~2,035.75 m ²
Capsule	6	10 m	~78.54 m ²	~471.24 m ²
Total	14	—	—	~2,506.99 m ²

Disk loading at a 150 t reference mass, if all eight main fans are credited, is approximately 0.72 kN/m². This is a preliminary geometric reference only. Final rotor design must close on power required, tip speed, noise, blade stress and control authority under single- and multi-module failure cases.

06. Main 18 m Lift-Fan System

The eight 18 m primary lift fans are the main source of aircraft lift and control authority. Each fan is conceived as an independent module with its own structural interface, drive system, monitoring and isolation capability. The large diameter is chosen to keep disk loading moderate for a heavy aircraft, but it introduces significant aeroelastic, manufacturing, transportation and acoustic challenges that must be retired by progressive testing.

Parameter	Baseline
Quantity	8
Diameter	18 m each
Role	Primary aircraft lift and control
Module independence	Required concept
RPM / Blade count / Solidity	TBD — subject to aeroelastic and acoustic sizing
Tip speed	TBD (acoustic and structural limits)
Motor / Inverter	TBD (electric drive)
Fire / thermal monitoring	Required
Controlled release	Required concept
Loss tolerance	Selected-module-loss capability required

Distributed lift and control authority with validated selected-module-loss capability is required. Local vibration, temperature, electrical and fire monitoring, safe isolation of power / fuel / fluid / data interfaces before emergency release, and prevention of failure propagation to adjacent modules are mandatory design objectives. Required engineering includes aerodynamic sizing, aeroelasticity, structural loads, vibration, thermal, acoustics and control allocation. The control system must be able to reallocate thrust and moment among remaining fans after one or more modules are limited or released, while keeping the aircraft within a defined recoverable envelope.

Manufacturing and logistics of 18 m class fans are non-trivial. Blade or complete fan transport, assembly, balancing, inspection and replacement procedures must be defined early. Composite blade design must address impact damage, fatigue, environmental effects and repairability. Tip-speed and noise constraints will strongly influence final RPM and blade geometry.

07. Six 10 m Safety Capsule Fans

The six 10 m fans are dedicated to the Safety Capsule after separation. They are not sized to lift the full aircraft mass; they are sized to control and recover the separated capsule mass under emergency conditions (including degraded power, wind and possible single-fan loss). Their placement around / below the capsule is intended to provide both vertical force and attitude control authority.

Parameter	Baseline
Quantity	6
Diameter	10 m each
Role	Emergency capsule lift / stabilization / descent control
Independent power / control	Required
RPM / Blade count	TBD
Loss tolerance	TBD by capsule dynamics analysis
Power source	Independent capsule battery / emergency supply

Exact thrust, power and control authority require a dedicated capsule aerodynamic model that includes ground / water effect, spray, wind and possible asymmetric fan loss. The capsule flight-control laws after separation must be simple, robust and independent of the main aircraft control computers. Deployment of parachutes (if used) must be coordinated with fan operation so that the two systems do not interfere destructively.

08. Propulsion & Turbogenerator Architecture

Provisional power chain: turbogenerator → protected segmented high-voltage distribution → inverter → electric motor → lift fan. The architecture is hybrid-electric in the sense that chemical energy (Jet-A) is converted to electrical power which then drives the fans. This allows physical separation of generation from the lift fans and supports modular isolation.

Item	Provisional Baseline
Turbogenerators	8
Rating (each)	~7 MW class
Installed generation	~56 MW
Fuel	Jet-A / Jet-A1
Final rating	TBD after losses and hover closure
Redundancy	Distributed / segmented
Thermal system	TBD — dedicated cooling required
Start / APU concept	TBD

The ~56 MW figure is an installed-generation target, not proof of hover capability. Final hover power must include aerodynamic, electrical, motor, inverter, thermal and control losses. Multiple generation channels, segmented HV buses, automatic fault isolation, independent capsule supply, controlled load shedding and emergency reserve, and physical / thermal separation of critical generation channels where practical are required. Generator failure cases must be analyzed for remaining power margin under hover and under emergency descent.

09. Electrical, Battery & Energy Systems

Aircraft battery provides transient power, bus stabilization and emergency support. Independent capsule battery supports life support, avionics, communications, emergency fans, parachutes, flotation and recovery systems. Battery capacity is TBD after mission-energy and emergency-duration closure. Historical concept reference (not frozen): 12–18 MWh aircraft battery buffer. Water contact must initiate a validated HV safe state. Electrical isolation must preserve capsule survival power.

High-voltage architecture must address insulation, arcing, fault detection, segregation of channels, water exposure and fire. The design should allow the capsule to remain powered even if the main aircraft HV system is deliberately shut down or has failed. Fault injection testing and water-immersion safe-state testing will be required for certification.

Fault	Desired Response
Single generator	Isolate and redistribute if margins permit
Bus fault	Segment and isolate
Inverter fault	Disable affected motor and reconfigure
Motor overtemperature	Limit / stop module
Insulation fault	Detect and isolate
Water contact	Validated water-safe electrical state
Capsule separation	Independent capsule power state
Multiple generator loss	Load shed + emergency power + land / separate

10. Fuel System

Parameter	Baseline
Fuel	Jet-A / Jet-A1
Initial mass	~30 t
Reference density	~0.8 kg/L
Reference volume	~37,500 L
Tank	Crashworthy / protected / compartmentalized
Automatic isolation	Required
Final fuel	TBD by mission simulation
Location philosophy	Separated from occupant survival volume

Fuel is protected from likely impact / fire paths and separated from the occupant survival volume. Automatic shutoff / isolation after credible impact or system failure is required. Interfaces to jettisonable modules must isolate before release. Tank design must address deformation, slosh, venting, thermal exposure, fire and water impact. Final fuel quantity must be based on complete mission burn plus reserve. Crashworthy tank technology and fire-suppression strategy must be consistent with the certification basis eventually chosen.

11. Thermal Management

Distributed generation creates substantial waste heat. Dedicated cooling is required for turbogenerators, motors, inverters, batteries and HV equipment. Thermal barriers prevent propagation to adjacent modules and the capsule. Water operation adds spray, ingestion and environmental constraints. The thermal system must be sized for worst credible high-power operation, ambient conditions, degraded cooling and blocked airflow.

Subsystem	Concern	Evidence Path
Turbogenerator	Combustion / exhaust / cooling	Thermal balance + test
Motor	Electrical losses	Model + dynamometer
Inverter	Switching losses	Thermal test
Battery	Charge / discharge heat	Cell / module tests
HV distribution	Conductor / joint losses	Electrical thermal test

Subsystem	Concern	Evidence Path
Fire boundary	Heat propagation	Fire tests
Capsule	Life-support heat rejection	Environmental test

Thermal design must also consider post-crash and post-water-impact conditions, where cooling airflow may be disrupted and residual heat or fire may threaten the capsule. Fire barriers and automatic isolation of high-temperature sources are therefore part of the thermal / fire architecture, not separate afterthoughts.

12. Structural Architecture & Materials

~70% CFRP target for primary structural mass. Metal / hybrid structure is used at high-load joints, hard points, module interfaces, impact structures and thermal / fire boundaries. Structure must address normal flight, asymmetric thrust, gusts, vibration, fatigue, damage tolerance, water impact, hard landing, capsize and jettison transients. Composite design must consider impact damage, environmental effects, joints, repair, fire and inspection. The central capsule creates a major load-path and crash-isolation interface.

Modular interfaces must transfer flight loads under normal conditions and allow controlled separation under emergency conditions. This dual requirement (strong retention + reliable release) is a classic design tension and will require careful analysis, testing and possibly pyrotechnic or mechanical release systems with multiple inhibits.

Load Case	Primary Concern
Normal hover / max mass	Global strength
Maximum climb	Propulsion + structure
Single-module loss	Asymmetric loads / control
Gust / maneuver	Global structure
Hard ground landing	Landing + capsule
Water landing	Hull / flotation
Water impact	Crash structure
Failed separation	Capsule mounts
Module jettison	Transient / global loads
Fire exposure	Thermal degradation
Fatigue	Life-cycle integrity
Transportation / ground handling	Support structure

13. Modular Airframe & Controlled Jettison

Replaceable / isolatable lift and power modules rather than a monolithic architecture. Modules contain mechanical, electrical, control, thermal and fire interfaces. Modules should be replaceable without dismantling the central capsule where practical. Interfaces carry flight loads and provide controlled emergency separation. Conceptual modularization allowance: ~8–14 t, subject to detailed design.

Intended Jettison Sequence

Detect abnormal condition → confirm severity → isolate fuel / HV / data / fluid as applicable → attempt suppression / containment → assess controllability → authorize release → verify clearance → reconfigure aircraft.

Jettison must not be triggered by a transient sensor anomaly. Validated inhibit logic must prevent unsafe release. Release is controlled clearance, not uncontrolled free fall. Released module trajectory must be analyzed against aircraft, terrain, people, vessels and infrastructure. Final actuator, energy source, timing and mechanical loads remain TBD. The decision logic (automatic vs crew-authorized) must be defined with clear human-factors and safety arguments.

14. Safety Capsule Architecture

The Safety Capsule is the defining feature of the GEN-3 concept. It is a central, detachable, buoyant, crashworthy volume that contains the occupants and carries its own emergency systems. The goal is that even if the rest of the aircraft is severely damaged or deliberately separated, the capsule can still protect and recover the people inside.

Parameter	Baseline
Location	Central
Type	Detachable / buoyant / crashworthy
Occupants	24–30 passengers + 2 crew target
Independent power	Yes
Life support	72 h target
Emergency fans	6 × 10 m
Parachutes	6–8 modular canopies
Water operation	Water landing / float
Egress	Multi-directional / inverted-state capable
Separation target time	~5–7 s from authorized trigger to primary separation
Crash isolation	Springs + dampers + progressive absorbers

The capsule is the central occupant survival volume and should be separated from highest-risk fuel and propulsion zones. Emergency systems remain functional after primary aircraft power loss. Capsule structure addresses water impact, hard landing and retained-airframe cases. Independent health monitoring and emergency controls are required.

14.1 Capsule Mounts & Crash Energy Management

If separation is delayed, fails or is not selected, the capsule-to-airframe interface should absorb and attenuate impact energy. Multiple attachment points, springs for controlled elastic response, dampers to dissipate energy, progressive absorbers for high-load crash events, mechanical travel limits and load paths that preserve capsule geometry are required. Mount count, stiffness, damping, crush force, stroke, ultimate load and occupant injury criteria remain TBD pending dynamic modeling and certification basis definition. The design must also consider the loads imposed on the capsule during normal flight and during asymmetric thrust cases so that the isolation system does not introduce excessive flexibility or vibration.

14.2 Capsule Separation & Emergency Propulsion

Target separation sequence: detection → decision → isolation → release → separation impulse → clearance → stabilization → recovery. Independent high-integrity emergency control path, mechanical release secure under normal flight loads and vibration, and validated inhibit logic are mandatory. Six 10 m fans provide lift and attitude control after separation. Power availability is independently assured. Control allocation should tolerate a validated single capsule-fan loss if margins permit. Water approach accounts for spray, wind, waves and ground effect. Parachute systems (if retained in the final architecture) must be coordinated with fan operation and with possible inverted or partially submerged states.

14.3 Life Support & Occupant Environment

72-hour life-support target is a design goal for post-separation or post-ditching survival until external rescue. This implies independent power, environmental control, communications, lighting, basic medical support and waste management appropriate to the occupant count. Detailed sizing of oxygen, CO₂ scrubbing, thermal comfort and battery endurance remains TBD and must be closed against the worst credible environmental conditions (including inverted or partially flooded states if those are retained as design cases).

15. Amphibious Operation, Water Impact & Egress

Controlled water takeoff and water landing, stable flotation, predictable attitude response, spray and water-ingestion protection, water-contact sensors and validated safe-state logic are design objectives. Initial environmental target: Sea State 3–4, subject to analysis and certification. Progressive water-impact energy absorption, potential sacrificial lower structures, crushable keel, deformable absorbers and segmented reinforced hull, plus capsule mounts for energy attenuation, support water-impact survival. Flotation remains available after credible single-zone damage. Fuel system isolates and minimizes release. Critical electrical equipment is protected or automatically isolated. Capsule remains structurally coherent after credible water-impact cases.

Self-righting or highly capsize-tolerant capsule is a target (low center of gravity, controlled buoyancy distribution, dedicated flotation chambers, geometry supporting recovery from inverted attitude). If complete self-righting cannot be guaranteed, inverted-state survival and egress remain required design objectives. Multiple emergency exits around the capsule support normal, side, tilted and inverted orientations. No single door is the only route in a credible capsize state. Doors operable after normal electrical power loss. Training and human-factors considerations for inverted egress must be addressed in the operational concept.

Water operation also affects maintenance, corrosion protection, sealing of critical volumes, and the definition of “safe to restart after water contact.” These operational and maintainability issues must be captured in later requirements and verification plans.

16. Flight Control, Avionics & Health Monitoring

Fly-by-wire architecture with conceptual triple-redundant primary flight-control computing, independent emergency capsule separation logic, distributed health monitoring, automatic thrust redistribution after selected module failure, automatic power redistribution after generator failure, fault annunciation and crew decision support, and deterministic, testable emergency modes. Sensor set includes inertial measurement, air data, fan speed / torque, motor / inverter temperature, structural vibration, fire / thermal state, fuel state, water-contact state, capsule orientation / separation and door / egress status.

The emergency capsule path must be independent of the primary flight-control computers so that a total primary-control failure does not prevent capsule separation and recovery. Software and hardware development assurance levels will be driven by the final hazard classification and certification basis.

Subsystem	Monitored Parameters	Response
Fan	RPM, vibration, temperature, torque	Limit / isolate / reconfigure
Motor	Current, temperature, insulation	Limit / isolate
Inverter	Temperature, voltage, faults	Isolate
Generator	Output, temperature, speed	Isolate / redistribute
Fuel	Level, pressure, leak, temperature	Isolate
Fire	Heat / smoke / flame / gas	Suppress / isolate / jettison
Structure	Strain / vibration (where instrumented)	Alert / restrict / land
Capsule	Pressure, power, orientation, water ingress	Independent survival mode
Water	Contact, attitude, flotation	Safe electrical / recovery

17. Preliminary Mass & Power Budgets

Mass and power remain the two most critical open numerical domains. The tables below capture the current conceptual allocation. They are not frozen and must be closed by progressive subsystem design, supplier data and integrated simulation.

Subsystem / Allowance	Status
Primary airframe	TBD; CFRP-dominant
8 main fan systems	TBD; major contributor
6 capsule fans	TBD
8 turbogenerators	TBD
HV / motors / inverters	TBD
Fuel	~30 t target
Fuel tanks / protection	TBD
Safety Capsule	TBD
Capsule mounts	~1.5–3 t conceptual
Water impact / flotation / self-righting	~2–4 t conceptual
Modular interfaces / jettison	Part of ~8–14 t allowance
Avionics / wiring / thermal / recovery	TBD
Margin	Required / program-defined
Dry mass region (preliminary)	~126–140 t
Dry + fuel planning region	~156–170 t possible upper region

At a 150 t reference mass, T/W 1.5 corresponds to approximately 225 metric-ton-force total thrust-equivalent; T/W 1.7 corresponds to approximately 255 metric-ton-force. These are sizing references only. Installed generation (~56 MW) is not equivalent to useful hover power. A validated model must include aerodynamic efficiency, induced power, profile power, motor / inverter / generator losses and thermal constraints. At 150 t, aircraft weight is approximately 1.471 MN. If all eight 18 m main fans are credited for primary hover, the average geometric disk loading is approximately 0.723 kN/m². This is a preliminary reference, not a final rotor design result.

Mass growth is one of the highest program risks. Every new safety feature, structural reinforcement, thermal system and battery addition must be tracked against the margin. Early and continuous mass-statement discipline is mandatory.

18. Safety Analysis Framework & Emergency Scenarios

The safety program should progress from qualitative hazards to formal functional hazard assessment (FHA), preliminary system safety assessment (PSSA) and quantitative reliability / risk evidence as required by the final certification basis. The layered-survivability philosophy is intended to keep the residual risk to occupants as low as practical even when individual systems fail.

Hazard	Consequence	Mitigation
Uncontained module fire	Module / structure threat	Containment + suppression + jettison
Multiple fan loss	Lift / control loss	Distributed propulsion + reconfiguration
Fuel rupture	Fire / explosion risk	Crashworthy tanks + isolation
Separation failure	High-energy impact exposure	Crash-isolated mounts
Parachute failure	High descent rate	Multiple canopies + fans
Capsule capsize	Delayed evacuation	Self-right / tolerance + egress
HV water exposure	Electrical hazard	Automatic isolation
Control computer fault	Loss of control	Redundancy + independent path
Jettison fault	Unintended separation	Retention + inhibit + monitoring
Mass growth	Performance / safety margin loss	Continuous mass control

Selected Emergency Scenario Matrix (excerpt)

Scenario	Initial Action	System Objective	Recovery
Single fan fire	Detect / isolate / suppress	Continue or jettison	Reconfigure / land
Uncontained module fire	Isolate + jettison	Controlled release	Emergency landing / capsule
One generator failure	Isolate	Redistribute	Continue / recover
Multiple generator faults	Load shed	Emergency power	Land / capsule
Capsule separation	Isolate / release	Independent mode	Fans / parachutes / water
Separation failure	Crash isolation	Protect capsule	Aircraft landing
Water impact	Fuel / HV isolation	Flotation	Egress / recovery
Capsize	Orientation detection	Self-right / stabilize	Egress
Primary control loss	Independent capsule path	Separate if required	Capsule recovery

Each scenario in the full matrix must eventually be supported by analysis, simulation and, where necessary, physical test. The matrix is a living document that will expand as the architecture is refined and as the certification basis is agreed.

19. Certification Strategy & Verification

The final certification category and exact certification basis cannot responsibly be frozen at this concept stage. FAA rotorcraft resources identify 14 CFR Part 27 (normal-category) and Part 29 (transport-category) as principal standards, with separate guidance. The actual category and basis must be established with the applicable authority. Novel features (large distributed fans, modular jettison, detachable capsule, amphibious recovery) will likely require special conditions or equivalent safety findings.

FAA aircraft-certification overview describes seven phases: Standards, Definition, Detail, Build, Test, Certification and Production. Upfront planning, establishment of certification basis, agreement on means of compliance, conformity, testing, type certification and production approval are required. Early authority engagement is strongly recommended. European (EASA) and other national authorities may be relevant depending on the intended market and production location.

Verification & Validation Highlights

- Material coupons (composite strength / fire / environment)
- Fan subscale (aerodynamics / vibration / aeroelasticity)
- Motor / inverter and turbogenerator bench testing
- Module ground rig (isolation / jettison)

- Fire module test (containment / suppression)
- Capsule drop and mount energy-absorption tests
- Separation rig (timing / clearance / dynamics)
- Parachute drop tests (if retained)
- Water impact, flotation and capsize tests
- Unmanned demonstrator → flight-test article → conforming prototype
- Integrated system safety assessment and flight-test program

Verification planning must start early so that critical tests (especially those involving large fans, crash isolation and water impact) can be designed, funded and scheduled. Some tests may require specialized facilities or international partners.

20. Development Roadmap (High-Level)

The roadmap is structured as a sequence of risk-reduction and capability-build phases. Each phase produces a defined output that either closes a major technical uncertainty or provides the next level of integrated demonstration. Funding, team size and facility requirements grow with the phases; early phases can be executed with a relatively small core team and selected specialist partners.

Phase	Focus	Key Output
0	Requirements / mission	Mass, power, performance closure
1	Aerodynamic sizing	Fan geometry / power model
2	Systems architecture	Interfaces / redundancy / safety
3	Subscale technology	Fans / motors / inverters / materials
4	Capsule demonstrator	Crash / separation / flotation
5	Module demonstrator	Fire / isolation / jettison
6	Integrated ground demonstrator	Power / control / safety
7	Unmanned flight demonstrator	Envelope / fault handling
8	Water demonstrator	Water landing / impact / recovery
9	Piloted prototype	Progressive flight
10	Certification program	Formal compliance
11	Production readiness	QMS / tooling / suppliers

Phases 0–3 are primarily analytical and subscale. Phases 4–6 introduce major physical demonstrators. Phases 7–9 move into flight and water. Phases 10–11 are the formal certification and production path. Parallel work on IP, supplier selection and early authority engagement should run alongside the technical phases.

21. Principal Engineering Risks

The risk register below captures the highest-priority technical risks at concept level. Each risk has a priority and a proposed retirement path. The register must be maintained and updated as analysis and testing progress. Residual risks that cannot be retired by design must be managed by operational limitations, training or other mitigations acceptable to the certification authority.

ID	Risk	Priority	Retirement Path
R-01	18 m fan structural / aeroelastic feasibility	High	Subscale + CFD + structural rig
R-02	~56 MW power / thermal integration	High	Power / thermal demonstrator
R-03	Mass growth	High	Subsystem closure + margin
R-04	Controlled module jettison	High	Ground dynamic tests
R-05	Capsule crash isolation	High	Drop / crash tests
R-06	Water-impact survival	High	Physical impact tests
R-07	Capsule self-righting	Med/High	Scale + full-scale tests
R-08	Certification treatment of novel features	High	Early authority engagement
R-09	Large composite fan manufacture	Med/High	Supplier demonstrator
R-10	Noise / vibration / acoustics	Med/High	Aeroacoustic tests
R-11	Thermal management under all conditions	High	Integrated power test
R-12	Asymmetric control after module loss	High	HIL + flight tests
R-13	Battery / energy storage for capsule endurance	Med	Cell / system tests
R-14	Human factors of inverted / water egress	Med	Mock-up + trials

22. Master Specification (Summary)

The master specification consolidates the principal numerical and architectural targets. It is the single place to look for the current “frozen concept” values. Any change to a master-specification entry should be controlled and impact-assessed.

Parameter	Baseline
Aircraft class	Heavy modular amphibious VTOL
Founder	Amirbahador Atai
Year / Revision	2026 / Rev 02 Full Expanded
Main lift	8 × ~18 m fans
Capsule lift	6 × ~10 m fans
Total fans	14
Main fan area (geometric)	~2,035.75 m ²
Capsule fan area (geometric)	~471.24 m ²
Combined geometric area	~2,506.99 m ²
MTOW	~150–160 t target (may grow)
Dry mass (preliminary)	~126–140 t
Fuel	~30 t preliminary target
Endurance	≥ 3 h + reserve
T/W	~1.5 design / ~1.7 max-emergency
Turbogenerators	8 × ~7 MW provisional
Installed generation	~56 MW provisional
Primary structure	~70% CFRP target
Safety Capsule	Central / detachable / buoyant / crashworthy
Occupants	24–30 + 2 crew target
Life support	72 h target
Separation time target	~5–7 s
Parachutes	6–8 modular canopies
Mounts	Springs + dampers + progressive absorbers
Main module release	Individual controlled jettison
Fire logic	Detect → Isolate → Suppress → Assess → Continue / Jettison
Water operation	Takeoff + landing + flotation + capsize tolerance
Egress	Multi-directional / inverted-state capable
Whole-aircraft parachute	TBD — dedicated feasibility study required
Cruise / range / RPM / detailed aero	TBD

23. Open Items / TBD Register (Selected)

The following list captures the major open numerical and architectural items that must be closed before the concept can be considered ready for detailed design freeze or significant investment in physical demonstrators. Each item should be assigned an owner, a target closure date and a method (analysis, test, supplier data, authority agreement).

- Final MTOW, dry mass, payload and mass margin
- Fuel mass / reserve and complete mission fuel burn
- Exact turbogenerator quantity / rating and installed power
- Battery capacity / chemistry for aircraft and capsule
- Fan RPM, blade count / geometry / solidity, tip speed and acoustic limits
- Motor and inverter ratings and efficiency maps
- Final capsule dimensions / mass / occupant capacity

- Mount count / stiffness / damping / stroke and crash-load criteria
- Separation actuator / timing / clearance and inhibit logic
- Capsule fan thrust / power and control authority after separation
- Parachute area / deployment sequence (if retained)
- Whole-aircraft parachute feasibility (if considered)
- Water operating envelope / sea-state limit
- Self-righting mechanism and inverted-state survival criteria
- Emergency door count / location and operability after power loss
- Certification category / basis / means of compliance
- Supplier selection for major systems
- Manufacturing process for large composite fans and capsule
- Final cost and schedule for each development phase

24. IP Architecture (Candidate Families) & Investor Readiness

Candidate invention families include: detachable central Safety Capsule; rapid capsule separation architecture; spring / damper / progressive energy-absorbing capsule attachment; six large emergency capsule fans; multi-canopy capsule recovery; amphibious capsule recovery; eight-module distributed lift architecture; fire-triggered module isolation and controlled jettison; controlled-trajectory module release; post-jettison thrust / power reconfiguration; distributed turbogenerator / electric propulsion; independent capsule power; water-impact survivability; self-righting / capsize-tolerant capsule; multi-orientation emergency egress; integrated layered aircraft / module / capsule safety architecture. These are candidate families only. They do not establish novelty, patentability, inventorship, ownership or freedom to operate. Formal prior-art searching and patent counsel are required.

Investor-facing strength should be demonstrated through coherent architecture tied to mechanisms, failure states, test programs, certification planning and risk retirement — not by claiming the aircraft is already production-ready. Clear mission, mass / power closure pathway, technology feasibility milestones, prototype strategy, IP ownership / filing strategy, certification pathway, supplier strategy, test plan, capital requirements by phase and measurable technical gates are required. Major numerical values remain targets until verified. Certification requires accepted means of compliance and evidence.

For licensing or sale of IP, the value will depend heavily on the depth of technical evidence (analysis and test), the clarity of ownership, the absence of blocking prior art, and the existence of a credible path to a certifiable product. A pure paper concept has limited value; a concept backed by subscale demonstrators and closed mass/power models has substantially higher value.

25. Manufacturing, Maintainability & Operational Considerations

Large composite fans, modular interfaces and a detachable capsule introduce manufacturing and logistics challenges that must be addressed early. Blade or complete-fan transport, assembly, balancing, inspection and replacement procedures must be defined. Modular architecture is intended to support module-level replacement without dismantling the capsule, which can improve maintainability if interfaces are well designed. Water operation imposes corrosion protection, sealing and post-water inspection requirements. Dispatch reliability targets and mean-time-between-unscheduled-removal figures must be established once the architecture is more mature.

Ground handling, hangar / dock requirements, and the ability to operate from austere or water sites without extensive ground equipment are part of the operational concept and should influence design choices for landing gear, flotation, and module access.

26. Design Integrity Statement

This Rev 02 Full Expanded Edition is intentionally comprehensive but remains a concept baseline. The architecture is detailed enough to guide the next engineering stage, while values dependent on detailed aerodynamics, structural dynamics, supplier data, certification criteria or testing remain targets / TBD. It is not evidence that the aircraft can presently fly, survive every stated scenario, obtain certification, or enter production.

The correct next step is controlled engineering closure: replace assumptions with calculations, calculations with validated models, critical models with tests, and maintain full traceability from each requirement to objective evidence. External certification context (FAA aircraft certification and rotorcraft guidance) is used only to frame the certification and verification sections; it does not constitute approval of this concept.

The founder and any future partners should treat this document as a living baseline. Every numerical update, architectural change or requirement change should be recorded with rationale and impact. Only through disciplined configuration control and progressive verification will the concept move from paper to a credible engineering program.

End of Document

GEN-3 SAFE HELICOPTER — Comprehensive Engineering Concept Design Description
Revision 02 – Full Expanded Edition • 2026 • Founder: Amirbahador Atai